PROBABILITY ON ONE PAGE

Poker hands from a pack

Five cards dealt from a well-shuffled pack, counted two ways, with the workings shown.

1 / THE RULE AND THE PICTURE

Count the favourable hands and divide by all the equally likely hands



full house: three 8s (navy) and two 3s (teal)

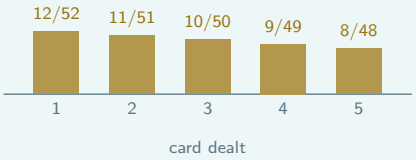
A hand is 5 cards in any order, so there are

$$\binom{52}{5} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5!} = 2\,598\,960$$

equally likely hands, and $\boxed{P = \dfrac{\text{favourable hands}}{\binom{52}{5}}}$

2 / FIVE PICTURE CARDS / TWO ROUTES, ONE ANSWER

$$\frac{12}{52} \times \frac{11}{51} \times \frac{10}{50} \times \frac{9}{49} \times \frac{8}{48} = \binom{12}{5} \Big/ \binom{52}{5}$$



Sequential. Each draw needs a picture card from those left: $\frac{12}{52}$, then $\frac{11}{51}$ given the first, then $\frac{10}{50}$, $\frac{9}{49}$, $\frac{8}{48}$, so $P = \dfrac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48} = \dfrac{95\,040}{311\,875\,200} \approx 3.05 \times 10^{-4}$.

Counting. Choose 5 of the 12 picture cards, $\binom{12}{5} = 792$ ways, from $2\,598\,960$ hands. The routes agree since both ordered counts carry the factor $5!$, which cancels.

$$P = \frac{792}{2\,598\,960} \approx 3.05 \times 10^{-4} \approx \frac{1}{3281}$$

3 / FULL HOUSE / THREE OF ONE RANK AND TWO OF ANOTHER

$$13 \times \binom{4}{3} \times 12 \times \binom{4}{2} = 3744 \textbf{ hands}$$

rank of the triple: 13
× its suits: $\binom{4}{3} = 4$
× rank of the pair: 12
× its suits: $\binom{4}{2} = 6$
= $13 \cdot 4 \cdot 12 \cdot 6 = 3744$

Counting. Choose the rank of the triple (13) and 3 of its 4 suits (4), then a *different* rank for the pair (12) and 2 of its 4 suits (6): $13 \cdot 4 \cdot 12 \cdot 6 = 3744$ hands, so $P = 3744/2\,598\,960 \approx 0.144\%$, about 1 in 694.

Sequential check. For the order XXXYY the first card is anything, then $\frac{3}{51} \cdot \frac{2}{50} \cdot \frac{48}{49} \cdot \frac{3}{48} = \frac{18}{124\,950}$, and the triple can sit in any of $\binom{5}{3} = 10$ positions: $10 \times \frac{18}{124\,950} = 0.001441$, the same value.

$$P = \frac{13\binom{4}{3} \cdot 12\binom{4}{2}}{\binom{52}{5}} = \frac{3744}{2\,598\,960} \approx 0.144\%$$

4 / EVERY HAND TYPE / THE COUNTS PARTITION ALL 2 598 960 HANDS

The same method, one line per hand

hand	workings	hands	probability	1 in
royal flush	4 suits ×1	4	0.00015%	649 740
straight flush	$10 \times 4 - 4$ (royal excluded)	36	0.0014%	72 193
four of a kind	$13 \times \binom{4}{4} \times 48$	624	0.024%	4165
full house	$13 \times \binom{4}{3} \times 12 \times \binom{4}{2}$	3744	0.144%	694
flush	$4\binom{13}{5} - 40$	5108	0.197%	509
straight	$10 \times 4^5 - 40$	10 200	0.392%	255
three of a kind	$13\binom{4}{3} \times \binom{12}{2} \times 4^2$	54 912	2.11%	47
two pair	$\binom{13}{2}\binom{4}{2}^2 \times 44$	123 552	4.75%	21
one pair	$13\binom{4}{2} \times \binom{12}{3} \times 4^3$	1 098 240	42.26%	2.4
high card	$\left(\binom{13}{5} - 10\right)(4^5 - 4)$	1 302 540	50.12%	2.0
total		2 598 960	100%	
five picture cards	$\binom{12}{5}$ (slide 2)	792	0.0305%	3281

ASIDE / WHY THE 5! CANCELS

Ordered deals number $52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 = 311\,875\,200$ and every hand appears $5! = 120$ times, so counting ordered favourable deals over ordered deals gives the same probability as counting hands.

ASIDE / CHECK IN PYTHON

`comb(52,5)` gives 2598960, `comb(12,5)/comb(52,5)` gives 3.047×10^{-4}, and `13*comb(4,3)*12*comb(4,2)` gives 3744.

Convention. A well-shuffled 52-card pack, five cards dealt, order ignored; straights include A-2-3-4-5 and 10-J-Q-K-A, and each hand is counted once, in its highest category.

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